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COSC 461  
Maniccam

The parameters used to obtain a minimum 70% accuracy are:

- 5 epochs
- 2 convolutional layers
- 96 convolutions
- 3x3 convolution size
- 2x2 pool size
- Maximum pool type
- 128 neurons per hidden layer
- Adam optimizer

```
import tensorflow as tf

data = tf.keras.datasets.cifar10

(training_images, training_labels), (test_images, test_labels) = data.load_data()

training_images = training_images.reshape(50000, 32, 32, 3)
test_images = test_images.reshape(10000, 32, 32, 3)
training_images = training_images/255
test_images = test_images/255

model = tf.keras.models.Sequential([
    tf.keras.layers.Conv2D(96, (3, 3), activation='relu', input_shape=(32, 32, 3)),
    tf.keras.layers.MaxPooling2D(2, 2),
    tf.keras.layers.Conv2D(96, (3, 3), activation='relu'),
    tf.keras.layers.MaxPooling2D(2, 2),
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(128, activation=tf.nn.relu),
    tf.keras.layers.Dense(10, activation=tf.nn.softmax)
])

model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

model.fit(training_images, training_labels, epochs=5)

model.evaluate(test_images, test_labels)

classifications = model.predict(test_images)

print(classifications[0])
print(test_labels[0])
```

```
Epoch 1/5
1563/1563 [=====] - 155s 98ms/step - loss: 1.3923 - accuracy: 0.5005
Epoch 2/5
1563/1563 [=====] - 164s 105ms/step - loss: 1.0276 - accuracy: 0.6421
Epoch 3/5
1563/1563 [=====] - 164s 105ms/step - loss: 0.8823 - accuracy: 0.6925
Epoch 4/5
1563/1563 [=====] - 160s 102ms/step - loss: 0.7830 - accuracy: 0.7282
Epoch 5/5
1563/1563 [=====] - 165s 105ms/step - loss: 0.6961 - accuracy: 0.7570
313/313 [=====] - 9s 28ms/step - loss: 0.8836 - accuracy: 0.7017
313/313 [=====] - 9s 28ms/step
[5.9994403e-03 1.4621406e-02 1.2817677e-02 5.6956100e-01 6.1839848e-04
 3.6602548e-01 2.4917943e-03 2.4407434e-03 2.5200527e-02 2.2353721e-04]
[3]
```